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CENTRE
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CANDIDATE
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CHEMISTRY

Paper 2

Grade 12

May 2020

2 hours

Candidates answer on the Question Paper.

Additional Materials: Calculator
Data Booklet

12CHEM/02

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

You may receive fewer marks if you do not show your working or do not use appropriate units where required.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 100.

Answer all questions in English.

For Examiner's Use

1	
2	
3	
4	
5	
6	
7	
8	
Total	

This document consists of **17** printed pages and **3** blank pages.

1 This question is about the use of balanced equations.

(a) (i) Copper reacts with concentrated sulfuric acid.

Copper(II) sulfate, sulfur dioxide and water are the products of the reaction.

Construct the balanced equation for this reaction.

..... [2]

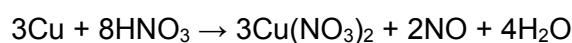
(ii) The mole is used in chemical calculations.

Define the term *mole*.

.....
..... [1]

(iii) A sample of 192 g of copper reacts with dilute nitric acid.

The equation for the reaction is shown.



Calculate the amount in moles of copper(II) ions / nitrate ions in the copper(II) nitrate formed in this reaction.

Show your working.

amount of copper(II) ion mol

amount of nitrate ion mol [3]

(b) The equation for a redox reaction is shown.



Write the ionic equation for this redox reaction indicating state symbols.

..... [2]

(c) The enthalpy change of solution of potassium manganate(VII) depends on the lattice energy and the enthalpy change of hydration.

(i) Define the term *enthalpy change of hydration*.

.....
..... [1]

(ii) Write equations for the enthalpy change of hydration for a potassium ion, K^+ , and a manganate(VII) ion, MnO_4^- .

Include state symbols in your equations.

enthalpy change of hydration for K^+
enthalpy change of hydration for MnO_4^- [2]

(iii) Draw the energy diagram for the enthalpy change of solution for potassium manganate(VII).

In your diagram, use lattice enthalpy of $KMnO_4$, enthalpy of hydration of $KMnO_4$ and enthalpy of solution of $KMnO_4$.

[3]

[Total: 14]

2 Elemental sulfur is found widespread in nature.

(a) State one source of sulfur in nature.

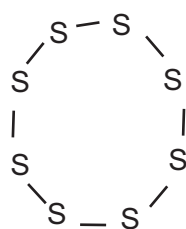
..... [1]

(b) Sulfur has several allotropes.

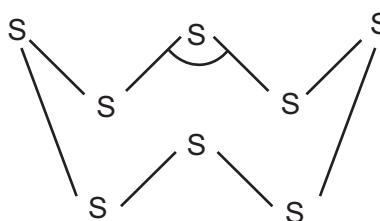
(i) Define the term *allotropes*.

..... [1]
.....

Two allotropes of sulfur are monoclinic sulfur and rhombic sulfur. Their structures are shown.



rhombic sulphur



monoclinic sulfur

(ii) Identify one similarity and one difference between these two allotropes.

similarity [1]
.....

difference [1]
.....

(c) In Kazakhstan sulfuric acid is produced from the waste gases from the extraction of non-ferrous metals.

(i) The flow chart shows the steps involved in the formation of sulfuric acid.



Write equations for each of the steps. Include state symbols.

step 1

step 2

step 3 [3]

(ii) Step 1 is an exothermic equilibrium reaction.

The compromise conditions for step 1 are 400 to 450 °C, 1 atmosphere pressure and a 1:1 mole ratio of SO₂ to O₂.

Explain how these three parameters affect the position of the equilibrium and the rate of reaction.

temperature

.....

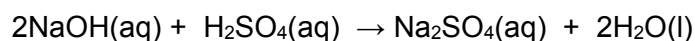
pressure

.....

mole ratio

..... [3]

(d) Sodium hydroxide reacts with sulfuric acid as shown in the equation.



A sample of 300 cm³ of 1.00 mol dm⁻³ NaOH(aq) is added to 100 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq).

Calculate the pH of the solution formed after the addition. Give your answer to **one** decimal place.

Show your working.

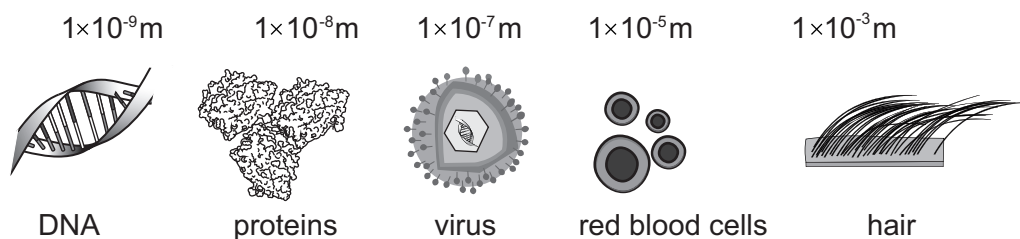
[5]

[Total: 15]

3 Nanotechnology is a modern area of chemistry.

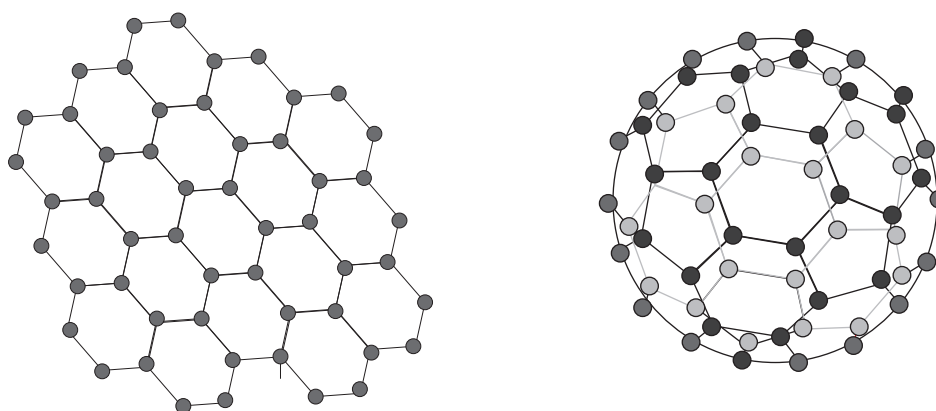
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(a) State which objects in the diagram have the size of a nanoparticle.



..... [1]

(b) Two nanoparticles of carbon are shown in the diagrams.



(i) Name and describe the structure of one of these nanoparticles.

name

description [2]

(ii) Carbon nanoparticles are used in water purification.

Describe one property of carbon nanoparticles which makes them suitable for this.

..... [1]

(c) Nanoparticles and polyethylene glycol (PEG) are used for targeted delivery of drugs.

(i) Describe the process of targeted delivery of drugs into the human body.

You may use a diagram for your answer.

.....

.....

.....

.....

..... [3]

(ii) Describe the advantages **and/or** disadvantages of using nanoparticles and of using PEG for the targeted delivery of drugs into the human body.

.....

.....

.....

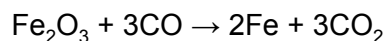
..... [3]

[Total: 10]

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- 4 Iron is extracted from iron(III) oxide, Fe_2O_3 , in a blast furnace.

Both carbon and carbon monoxide reduces iron(III) oxide in a blast furnace.



Some thermochemical data is shown in the table.

	$\text{Fe}_2\text{O}_3(\text{s})$	$\text{Fe}(\text{s})$	$\text{C}(\text{s})$	$\text{CO}(\text{g})$	$\text{CO}_2(\text{g})$
standard enthalpy of formation, $\Delta H_f^\theta / \text{kJ mol}^{-1}$	-824.2	0	0	-110.5	-393.5
standard entropy, $S^\theta / \text{JK}^{-1} \text{mol}^{-1}$	87.4	27.3	5.7	197.6	213.6

- (a) Calculate the standard enthalpy change, ΔH^θ , and standard entropy change, ΔS^θ , for the reduction of iron(III) oxide by carbon monoxide.

Show your working.

$$\Delta H^\theta = \dots\dots\dots \text{kJ mol}^{-1}$$

$$\Delta S^\theta = \dots\dots\dots \text{JK}^{-1} \text{mol}^{-1} \quad [4]$$

- (b) The equation for the reduction of iron(III) oxide by carbon is shown.



The standard entropy change, ΔS^θ , for this reaction is $+542.6 \text{ JK}^{-1} \text{mol}^{-1}$

Calculate the minimum temperature needed for this reaction to become spontaneous.

Quote your answer to **three** significant figures.

Minimum temperature K [2]

- (c) The reduction of iron(III) oxide by carbon monoxide is spontaneous at room temperature.

Use ideas about enthalpy change and entropy change to explain the decrease of the minimum temperature in carbon monoxide reduction in comparison to the results calculated in (b).

.....

.....

.....

..... [2]

- (d) (i) Carbon monoxide reacts quickly with haemoglobin. Describe the function of haemoglobin in the human body.

.....

..... [1]

- (ii) Explain how carbon monoxide prevents haemoglobin to function.

.....

.....

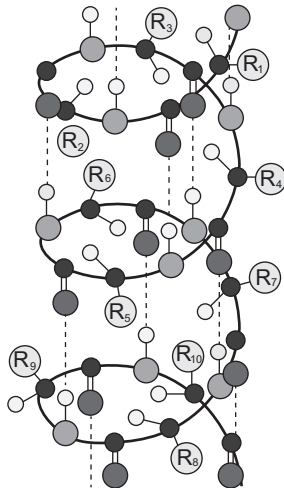
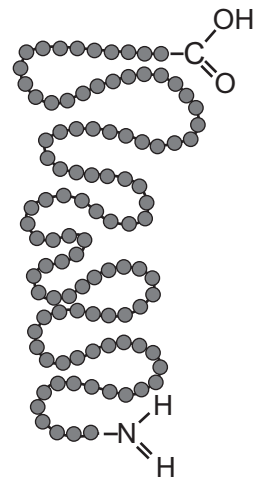
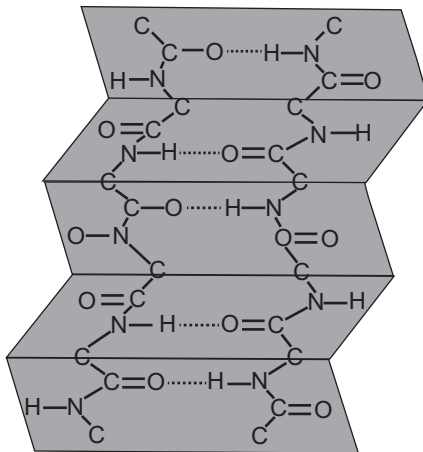
.....

..... [2]

[Total: 11]

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5 Some protein structures are shown in the diagrams.

**G****H****K****L**

(a) (i) Identify the secondary structures shown in **G** and in **K**.

G

K [2]

(ii) Describe how the structures in **G** and **K** are formed.

.....

.....

.....

..... [2]

(iii) Describe a primary structure and its chemical bonding.

.....

.....

.....

..... [3]

(b) Define the term *denaturation*.

.....

..... [1]

(c) State **two** different methods of denaturing a protein.

For each method give an example of this type of denaturation.

method 1

example

method 2

example [2]

(d) A protein contains a disulfide bridge.

Explain how copper(II) ions can change the properties of this protein.

.....

.....

.....

..... [2]

[Total: 12]

6 (a) Iron can form several complex ions such as $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$.

(i) Write the **full** electronic configuration for Fe^{3+} .

Fe^{3+} [1]

(ii) State **and** explain which complex ion, $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ or $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, shows the greater acidic character.

Complex

Explanation

.....

..... [2]

(b) Aqueous carbonate ions react with aqueous $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ in a precipitation reaction.

(i) Construct the balanced equation, including state symbols, for this reaction.

..... [2]

(ii) Describe the observations of this reaction.

..... [1]

(c) Aqueous carbonate ions can also react with aqueous $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ in a precipitation reaction.

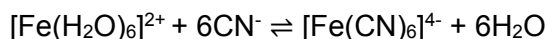
(i) Construct the balanced equation for this reaction.

..... [1]

(ii) Describe the observations of this reaction.

..... [1]

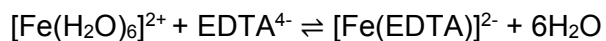
(d) The complex ion $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ reacts with cyanide ions as shown:



Write the expression for the stability constant, K_{stab} , of the equilibrium reaction.

..... [1]

(e) The complex ion $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ reacts with EDTA^{4-} as shown:



(i) Suggest why $[\text{Fe}(\text{EDTA})]^{2-}$ is more stable than $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$.

.....
 [1]

(ii) Use ideas about entropy of the system and Gibbs free energy to suggest why the reaction of $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ with EDTA^{4-} is difficult to reverse.

.....

 [3]

[Total: 13]

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7 Alcohols are widely used for some industrial processes.

(a) (i) An alcohol has the molecular formula $C_4H_{10}O$.

Draw the structure of three isomers of $C_4H_{10}O$ that are alcohols.

Name each of the isomers you draw.

[3]

(ii) Alcohols can undergo reactions involving the loss of water.

Name this type of reaction.

..... [1]

(iii) State the conditions for this reaction.

..... [1]

(b) (i) In industry, ethanol is produced by fermentation and by the hydration of ethene.

Put ticks (✓) in the boxes to show the correct information about fermentation and hydration.

	higher rate of reaction	more expensive equipment	lower atom economy
fermentation			
hydration			

[2]

(ii) Bioethanol is a biofuel made from glucose.

Use chemical equations to explain why bioethanol is a carbon-neutral fuel.

.....

.....

.....

.....

.....

.....[3]

(c) Potassium dichromate(IV) is heated under reflux with an excess of ethanol until an obvious change in colour is detected.

(i) Describe the colour change observed during the reaction.

..... [1]

(ii) Write the organic product of this reaction.

..... [1]

[Total: 12]

8 Aluminium is in Group 13 of the Periodic Table.

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(a) (i) Draw a labelled diagram of the metallic structure of aluminium.

[3]

(ii) Explain why aluminium has a high melting point.

.....
.....
.....
.....
..... [2]

(iii) Explain why aluminium has a high thermal conductivity.

.....
.....
.....
.....
..... [2]

(b) Write an equation to represent the third ionisation energy of aluminium.

Include state symbols in your equation.

..... [2]

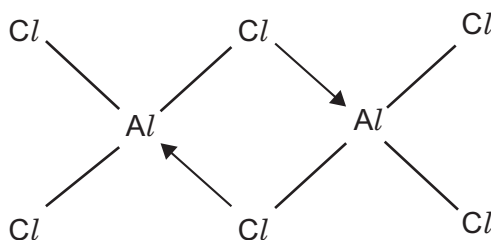
(c) Draw the 'dot-and-cross' diagram for a molecule of aluminium chloride, $AlCl_3$.

Include only the outer shell electrons.

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[2]

(d) At low temperatures aluminium chloride forms a dimer with the formula Al_2Cl_6 .



Suggest why $AlCl_3$ molecules form the dimer Al_2Cl_6 .

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.....

.....

.....

..... [2]

[Total: 13]

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